

SHAHERYAR AMIR

AI/ML Engineer | Software Developer | Computer Science Student

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PROFESSIONAL SUMMARY

Computer Science student at FAST-NUCES with hands-on experience building and deploying machine learning and NLP systems across two industry internships. Skilled in supervised learning, feature engineering, anomaly detection, retrieval-augmented generation (RAG), and full-stack development in Python and C++, with a track record of translating technical results into clear insights for technical and non-technical audiences alike.

EXPERIENCE

AI / ML Intern | Elevvo Pathways

Jun – Jul 2025

- Trained, tuned, and evaluated supervised machine learning models (scikit-learn, pandas) on real-world datasets, covering the full model development lifecycle from preprocessing to evaluation.
- Designed and implemented feature engineering pipelines and anomaly detection algorithms, improving model accuracy and robustness on noisy production-style data.
- Presented analytical findings and model behavior to cross-functional stakeholders, translating technical results into actionable, non-technical insights.

Web Development Intern | Digital Empowerment Network

Jul – Sep 2025

- Built and improved full-stack web projects during a remote internship, applying front-end and back-end development practices.
- Delivered consistent, high-quality contributions across assigned development tasks, demonstrating proactive ownership and reliability. (Intern ID: STU-WD-251-813)

PROJECTS

Medora Health App — Python, NLP, RAG, scikit-learn

- Designed an AI-driven healthcare platform enabling patients to log symptoms, receive NLP-based preliminary health assessments, book appointments, and manage personalized medicine reminders.
- Architected a multi-role system (User/Doctor/Admin) with role-based access, a doctor-verification workflow for AI predictions, and a retrieval-augmented QA layer grounded in verified medical sources.
- Incorporated NLP-based medical report simplification and symptom-to-disease mapping using a structured medical knowledge base.

AI-Powered Multimodal Study Assistant (RAG Pipeline) — In Progress — Python, Ollama, ChromaDB, Hugging Face, Whisper

- Building an end-to-end local RAG system that ingests PDFs, lecture videos, and images, indexing them in ChromaDB with multimodal embeddings (MiniLM, CLIP).
- Implementing hybrid retrieval (vector search + BM25) with cross-encoder reranking to boost precision, paired with a local LLM (Ollama) for citation-grounded answer generation.
- Developing an adaptive learning layer tracking quiz performance and computing per-topic mastery scores to recommend targeted review material.

FraudX — Python, scikit-learn, Anomaly Detection

- Built a real-time transaction monitoring system using ML-based anomaly detection to flag suspicious activity.
- Developed a live risk dashboard with automated alerts to support fraud analysts in reviewing flagged transactions.

SpaceShooter — C++, SFML

- Developed a 2D space shooter game featuring real-time collision detection, an enemy AI state machine, dynamic difficulty scaling, and a persistent high-score system.

SocialMediaApp — C++, OOP

- Built a console-based social network simulation with user profiles, bidirectional friend graphs, post creation, and a personalized news feed.

TECHNICAL SKILLS

Languages: Python, C++, x86 Assembly

AI / ML: Machine Learning, NLP, RAG, Anomaly Detection, Feature Engineering, Model Evaluation, LLMs (Ollama, Hugging Face)

Tools & Libraries: scikit-learn, pandas, ChromaDB, Whisper, Git, Jupyter, SFML

Concepts: Object-Oriented Programming, Data Structures & Algorithms, Software Architecture

Soft Skills: Problem Solving, Technical Communication, Team Collaboration, Analytical Thinking, Research & Presentation

EDUCATION

BS Computer Science | FAST-NUCES, Lahore — National University of Computer & Emerging Sciences | 2023 – Present

Intermediate — FSc Pre-Engineering | Punjab Group of Colleges, Lahore

LEADERSHIP & INVOLVEMENT

- Officer, Softec PC — Organized and coordinated activities for FAST-NUCES's flagship annual software exhibition.